

between elements of sets such as those between roll numbers of students in a class and their names, industries and their telephone numbers, employees in an organization and their salaries occur. Relations can be used to solve problems such as producing a useful way to store information in computer databases.

Definition

When A and B are sets, a subset R of cartesian product $A \times B$ is called a Binary relation from A to B viz, if R is Binary relation from A to B , R is a set of ordered pair (a, b) , where $a \in A$ and $b \in B$, when $(a, b) \in R$.

When $(a, b) \in R$, we use the notation $a R b$ and read it as " a is related to b by R " if $(a, b) \in R$ it is denoted as $a \in R b$.

Note: If R is relation from a set A to itself viz. If R is subset of $A \times A$, then R is called a relation on the set A .

The set $\{a \in A / a R b \text{ for some } b \in B\}$ is called the domain of R and denoted by $D(R)$.

The set $\{b \in B / a R b \text{ for some } a \in A\}$ is called the range of R and denoted by $R(Z)$.

Types of Relation

The Relation R on set A is called universal relation, if $R = A \times A$,

for example if $A = \{1, 2, 3\}$ then $R = A \times A$

$= \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$ is the universal relation on A .

- **Void Relation:** A relation R on set A is called void Relation, if R is the null set.

For example if $A = \{3, 4, 5\}$ and R is defined as $a R b$ if and only if $a + b > 10$, then R is the null set since no element in $A \times A$ satisfies the given condition.

- **Identity Relation:** A relation R on set A is called identity Relation, if $R = \{(a, a) / a \in A\}$ and is denoted by I_A .

For example if $A = \{1, 2, 3\}$ and then $R = \{(1, 1), (2, 2), (3, 3)\}$ is the identity relation on A .

- **Inverse Relation:** When R is any relation from a set A to a set B , the inverse of R , denoted by R^{-1} is the relation from B to A which consist of those ordered pairs (b, a) by interchanging the elements of the ordered pairs in R .

$$R = \{(a, b)\}$$

then

$$R^{-1} = \{(b, a)\}.$$

2.9. PROPERTIES OF RELATION

- A relation R on a set A is said to be reflexive, if $a R a$ for every $a \in A$.

i.e., of $(a, a) \in R$ for every $a \in A$.

For example: If R is the Relation on $A = \{1, 2, 3\}$ defined by $(a, b) \in R$ if $a \leq b$ where $a, b \in A$, then $R = \{(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$

Now R is Reflexive, since each element of A is related to itself as $(1, 1), (2, 2), (3, 3)$ are members in R .

- (ii) A relation R on a set A is said to be symmetric, if whenever $a R b$ then $b R a$ i.e., if whenever $(a, b) \in R$ then $(b, a) \in R$.

Thus Relation R on A is not symmetric if there exist $a, b \in A$ such that $(a, b) \in R$ but $(b, a) \notin R$.

- (iii) A relation R on A is said to be antisymmetric whenever (a, b) and $(b, a) \in R$ then $a = b$. If there exist $a, b \in A$ such that (a, b) and $(b, a) \in R$ but $a \neq b$.

Then R is not antisymmetric.

For Example: $R = \{(1, 3), (3, 1), (2, 3)\}$ is neither symmetric nor antisymmetric.

But the Relation $S = \{(1, 1), (2, 2)\}$ is both symmetric and antisymmetric.

- (iv) A relation R on a set A is said to be transitive, if whenever (a, b) and $(b, c) \in R$ then $(a, c) \in R$.

Thus if there exist $a, b, c \in A$ such that (a, b) and $(b, c) \in R$ but $(a, c) \notin R$ then R is not transitive.

- (v) A relation R on a set A is equivalence relation, if R is Reflexive, symmetric and transitive.

vz. R is equivalence relation on set A if it has the following three

1. $a R a$ for every $a \in A$.
2. if $a R b$ then $b R a$.
3. if $a R b$ and $b R c$ then $a R c$.

- (vi) A relation R on a set A is called partial ordering or partial order relation, if R is reflexive, antisymmetric and transitive.

R is a partial order relation on A if it has the following three properties:

- (a) $a R a$ for every $a \in A$.
- (b) $a R b$ and $b R a \Rightarrow a = b$
- (c) $a R b$ and $b R c \Rightarrow a R c$.

2.10. POSET

A set A together with a partial order relation R is called partially ordered set or poset. For example, the greater than or equal to relation is a partial ordering on the set. In a POSET the notation $a \leq b$ (or equivalently $a < b$) denotes $(a, b) \in R$. $a \leq b$ is read as " a precedes b " or " b succeeds a ".

Similarly, when $\{P, \leq\}$ is a poset, an element $a \in P$ is called maximal member of P , if there is no element $b \in P$. Such that $a < b$ (viz. strictly precedes b).

Similarly, an element $a \in P$ is called minimal member of P , if there is no element $b \in P$ such that $b < a$.

- if there exists an element $a \in P$ such that $b \leq a$ for all $b \in P$, then a is called the greatest member of the poset $\{P, \leq\}$.
- if there exists an element $a \in P$, such that $a \leq b$ for all $b \in P$, then a is called the least member of the poset $\{P, \leq\}$.

Properties of Relations;

Reflexive: A relation R on a set A is reflexive if $(a,a) \in R$ for all $a \in A$.

Symmetric, Asymmetric and Antisymmetric!

A relation R on a set A is symmetric if whenever aRb , then bRa . It then follows that R is not symmetric if we have some a and $b \in A$ with aRb , but $b \not R a$.

diagonal not allowed
A relation R on a set A is asymmetric if whenever aRb , then $b \not R a$. It then follows that R is not asymmetric if we have some a and $b \in A$ with both aRb and bRa .

allow diagonal elements.
A relation R on a set A is antisymmetric if whenever aRb and bRa then $a=b$. The contrapositive of this definition is that R is antisymmetric if whenever $a \neq b$ then $a \not R b$ or $b \not R a$. It follows that R is not antisymmetric if we have a and b in A , $a \neq b$ and both aRb and bRa .

Transitive Relations:

We say that a relation R on a set A is transitive if whenever aRb and bRc then aRc . It is often convenient to say what it means for a relation to be not transitive. A relation R on A is not transitive if there exist a, b and c in A so that aRb and bRc but $a \not R c$. If such a, b and c does not exist, then R is transitive.

(a) Consider the following ^{sets} five relations on the set $A = \{1, 2, 3, 4\}$

$$R_1 = \{(1, 2), (2, 1), (2, 3), (1, 3), (4, 4)\}$$

$$R_2 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$$

$$R_3 = \{(1, 3), (2, 1)\}$$

$$R_4 = \phi \text{ the empty relation}$$

$$R_5 = A \times A, \text{ universal relation}$$

determine which of the relations are reflexive, symmetric, antisymmetric & transitive.

Reflexive — R_1, R_2, R_5

Symmetric — R_2, R_5

Antisymmetric — R_5 , $R_4?$

Transitive — R_2, R_5, R_1 , $R_4?$

Equivalence Relation: A relation R on a set 'A' is said to be equivalence if R is.

(a). Reflexive (b) Symmetric (c). Transitive.

$$A = \{1, 2, 3\}$$

$\times R_1 = \{ \} \rightarrow R - \times$ ^{No (1,1) (2,2) (3,3)}
selection by elimination.

$\checkmark R_2 = \{ (1,1) (2,2) (3,3) \}$ $\begin{matrix} R - \checkmark \\ S - \checkmark \\ T - \checkmark \end{matrix}$
smallest pair. No pair to merge to answer

$\times R_3 = \{ (1,1) (2,2) (3,3) (2,1) \}$ $\begin{matrix} R - \checkmark \\ S - \times \end{matrix}$ *No (1,2) for (2,1)*

$\times R_4 = \{ (1,1) (1,3) (2,1) (3,1) \}$ $\begin{matrix} R - \times \\ S - \checkmark \\ T - \checkmark \end{matrix}$ *No diagonal pair*

$\checkmark R_5 = \{ (1,1) (2,2) (3,3) (1,2) (1,3) (2,1) (3,1) \}$ $\begin{matrix} R - \checkmark \\ S - \checkmark \\ T - \checkmark \end{matrix}$

$\times R_6 = \{ (1,1) (1,2) (2,1) (2,3) (3,1) (3,2) (3,3) \}$ $\begin{matrix} R - \times \\ S - \checkmark \\ T - \checkmark \end{matrix}$ *No (2,2)*

$\checkmark R_7 = A \times A$
Largest set.

Partial order Relation (Poset): A Relation R on a set 'A' is set to be Poset, if it is.

(a) Reflexive (b). Anti-symmetric (c). Transitive.

$$A = \{1, 2, 3\}$$

$$\times R_1 = \{ \} \quad \times \text{ not Reflexive.}$$

$$\checkmark R_2 = \{(1,1) (2,2) (3,3)\} \quad \begin{array}{l} R - \checkmark \\ AS - \checkmark \\ T - \checkmark \end{array}$$

$$\bullet \quad \times R_3 = \{(1,1) (2,2) (3,3) (1,2) (2,1)\}$$

Symmetry.
↓
Not Anti-Symmetric.

$$\checkmark R_4 = \{(1,1) (2,2) (3,3) (1,3) (2,3)\} \quad \begin{array}{l} R - \checkmark \\ AS - \checkmark \\ T - \checkmark \end{array}$$

$$\times R_5 = \{(1,1) (1,2) (2,3) (1,3)\} \quad R - \times$$

$$\times R_6 = A \times A \text{ — will have all the pairs & be symmetric so can not be Poset.}$$

$$(3,4) \quad \times \text{ No Poset} \quad \text{integer.}$$

$$R_1 = \{(a,b) \mid a, b \in \mathbb{Z}, a < b\}$$

No integer like $3 < 3$
so No diagonal element, No Reflexive, so No Poset.

$$\checkmark \text{Ref} - 3 < 3$$

$$\checkmark R_2 = \{(a,b) \mid a, b \in \mathbb{Z}, a \leq b\}$$

$$\checkmark \text{AS} - (3,4) \not\Rightarrow (4,3) \quad \times$$

yes Poset.

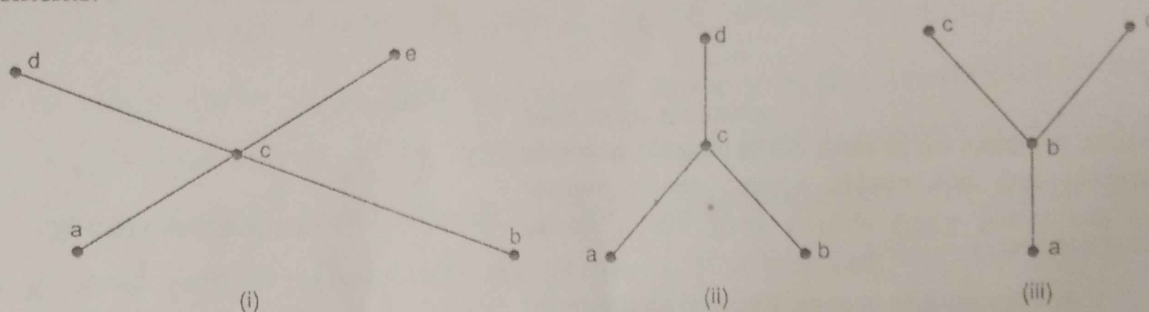
$$T - (1,4) (4,10) (1,10)$$

Note:

The maximal, minimal, the greatest and least members of a poset can be easily identified using the hasse diagram of the poset. They are the top and bottom elements in the diagram.

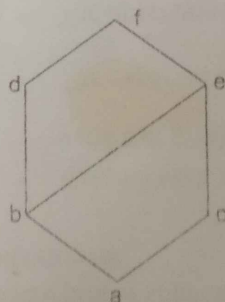
for the poset with Hasse diagram (i), a and b are minimal elements and d and e are maximal elements, but the poset has neither the greatest nor the least element for the poset with Hasse diagram (ii), a and b are minimal elements and d is the greatest element (also the only maximal element) there is no least element.

for the poset with Hasse diagram (iii) a is the least element and c and d are maximal elements.



2.11. HASSE DIAGRAM

When A is a subset of a poset $\{P, \leq\}$ and u is an element of P such that $a \leq u$ for all elements $a \in A$, then u is called upper bound of A . Similarly if l is an element of P such that $l \leq a$ for all elements $a \in A$, then l is called a lower bound of A .



Note: The upper and lower bound of a subset are not unique.

The element x is called L U B (least upper bound) or supremum of the subset A of a poset $\{P, \leq\}$, if x is an upper bound that is less than every other upper bound of A .

Similarly y is called greatest lower bound (GLB) or infimum of the subset A of a poset $\{P, \leq\}$ if y is a lower bound that is greater than every other lower bound of A .

Note: The L U B and GLB of a subset of a poset, if they exist, are unique.

For Example: Upper bound of the subset.

$\{a, b, c\}$ are c and b .

$[d$ is not the upper bound, since c is not Related to $d]$.

and L U B of $\{a, b, c\}$ is e .

HASSE DIAGRAMS:

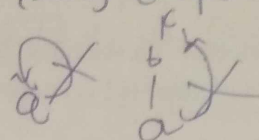
in mathematical discipline known as order theory, A Hasse diagram named after Helmut Hasse (1898-1979) is a simple picture of a finite partial order set, forming a drawing of the transitive reduction of the partial order. Concretely, one represent each element of S as a vertex on the page and draws a line segment or curve that goes to upward from x and y if $x < y$ and there is no z such that $x < z < y$.

in this case, we say y covers x or y is an immediate successor of x . furthermore it is required that the vertices are positioned in such a way that each curve meets exactly two vertices, its two end points. Any such diagram (given that the vertices are labeled) uniquely determine a partial order, that any partial order has a unique transitive reduction but there are many possible placements of elements in the plane, resulting in different Hasse diagrams for a given order that may have widely varying appearances.

Hasse diagram:- $(P, \leq)$

because relation is for
& Reflexive, must have self loop
for every element
↓
so for transitive

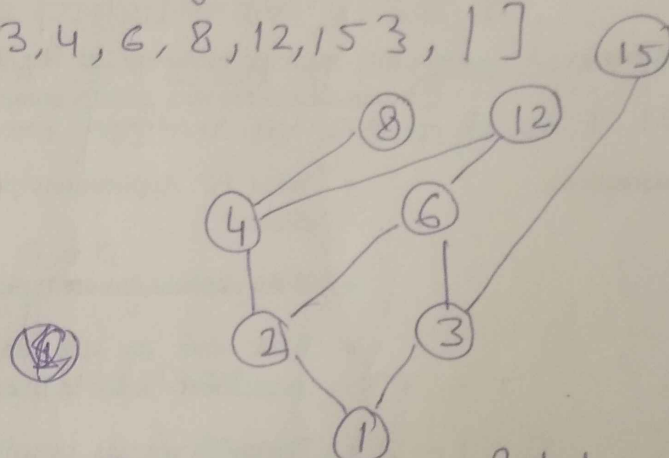
- 1) Create a vertex for element of your domain.
- 2) if aRb then draw an edge from a to b ($\begin{smallmatrix} b \\ a \end{smallmatrix}$)
- 3) Remove self loops. transitive edges.



Ex. Construct Hasse diagram for.

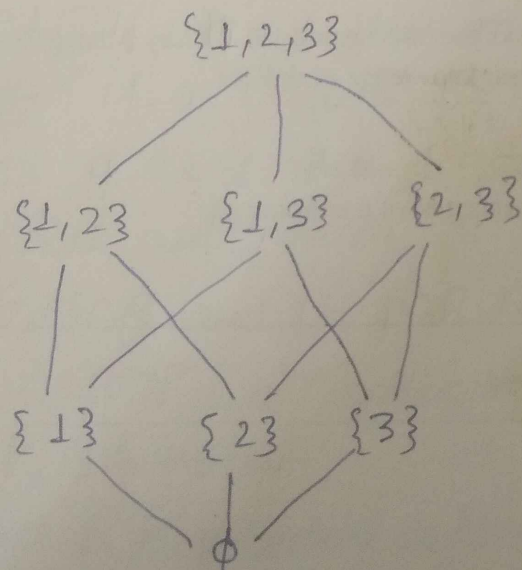
$\{1, 2, 3, 4, 6, 8, 12, 15\}, |$

given Natural Nds &
relation /



Ex. $A = \{1, 2, 3\}$ for Powerset of A & Relation $\subseteq [P(A), \subseteq]$

$P(A) = \{ \emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\} \}$



Ex: Prove that relation $\subseteq$ of set inclusion is a partial ordering on any collection of sets.

Sol: let A and B are two sets

we know $A \subseteq A \therefore (A, A) \in R$ i.e. R is reflexive

if $A \subseteq B$ and $B \subseteq A$ then $A = B$ i.e. R is antisymmetric

if $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$ i.e. $(A, B) \in R$ and $(B, C) \in R$
then $(A, C) \in R \therefore R$ is transitive.

Hence R is a partial ordering Relation.

Ex: if R is the relation on the set of integers such that $(a, b) \in R$ iff $b = a^m$ for some positive integer m then R is a partial ordering Relation.

Sol: - we know $a = a^1$

$\therefore (a, a) \in R$

So R is reflexive Relation

let $(a, b) \in R$ then $b = a^m$ and $(b, a) \in R$

then $a = b^n$ i.e. $a = (a^m)^n = a^{mn}$

that means $a = a^{mn}$ so $mn = 1$ or $a = 1$ or $a = -1$

Case 1: if $mn = 1$ then $m = 1$ and $n = 1 \therefore a = b$

Case 2: if $a = 1$ then $b = 1^m = 1 = a$

Case 3: if $a = -1$ then $b = -1$ so $a = b$

$\therefore R$ is antisymmetric

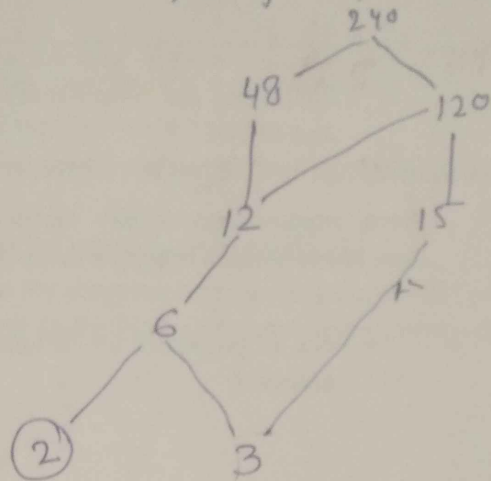
let $(a, b) \in R$ and $(b, c) \in R$ i.e. $b = a^m$ and $c = b^n$

$\therefore c = (a^m)^n = a^{mn}$ where mn is integer

so R is transitive

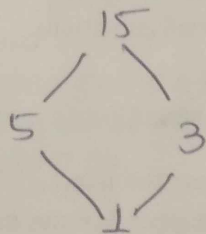
Hence R is a partial ordering Relation.

Ex. $[\{2, 3, 6, 12, 15, 48, 120, 240\}, 1]$ relation No's & relst and div

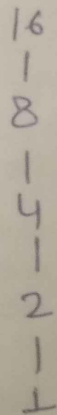


Ex: 9. Let D_m denotes the positive divisors of moderated by divisibility. Draw the Hasse diagrams of.

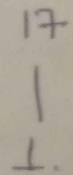
(a). D_{15}



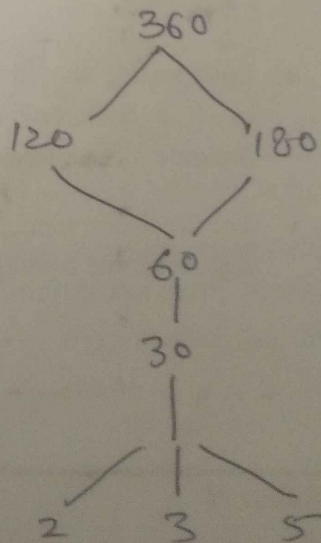
(b). D_{16}



(c). D_{17}



Ex. $\{2, 3, 5, 30, 60, 120, 180, 360\}$



(c) R_3 is reflexive, symmetric and transitive.

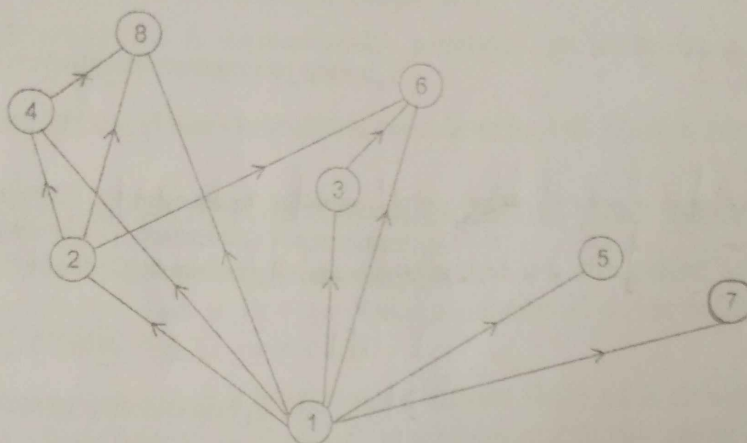
$\therefore R_3$ is an equivalence relation.

(d) R_4 is reflexive and symmetric but not transitive since $(1, 3)$ and $(3, 2) \in R_4$ but $(1, 2) \notin R_4$.

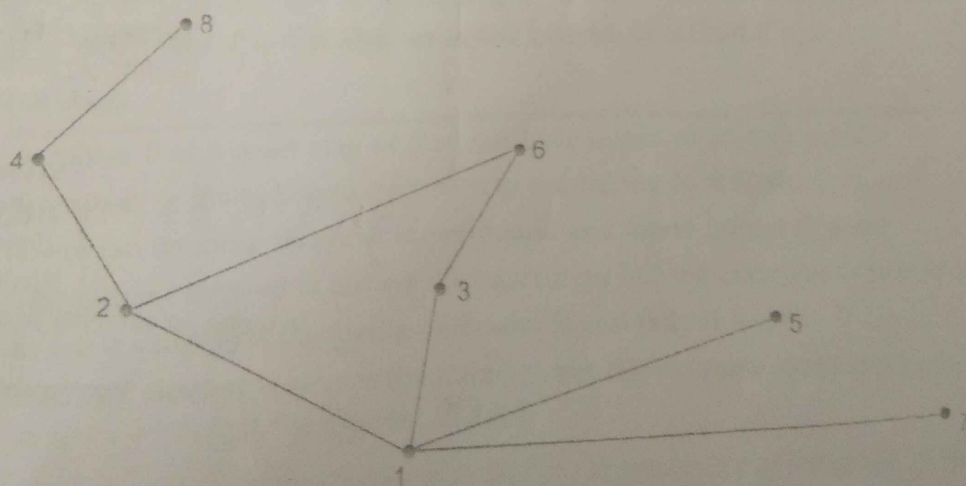
$\therefore R_4$ is not an equivalence relation.

Example 4. Draw the diagram representing the partial ordering $\{(a, b)/a \text{ divides } b\}$ on the set $\{1, 3, 4, 5, 6, 7, 8\}$. Reduce it to the Hasse diagram representing the given partial ordering.

Solution.



Deleting all the loops at the vertices, deleting all the edges occurring due to transitivity, arranging all the edges at point upward and deleting all arrows, we get the corresponding Hasse diagram.



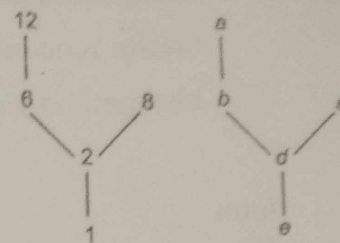
Example 5. Draw the Hasse diagram representing partial ordering $\{(A, B)/A \subseteq B\}$ on the power set $P(S)$, where $S = \{a, b, c\}$. Find the maximal, minimal, greatest and least elements of the poset.

Solution.

$$P(S) = \{\{\phi\}, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$$

Solution. The similarity mapping preserves the order of the initial set X and is one-to-one and onto. Thus the mapping can be viewed simply as a relabeling of the vertices in the Hasse diagram of the initial set X .

The Hasse diagram for X and Y are as follows :



Hence, if a poset A is isomorphic to a poset B and $f: A \rightarrow B$ is a similarity mapping, then

1. An element $a \in A$ is a first (last, minimal or maximal) element of A iff $f(a)$ is a first (last, minimal or maximal) element of B .
2. An element $a \in A$ immediately precedes an element $a' \in A$ that is $a << a'$ iff $f(a) << f(a')$.
3. An element $a \in A$ has r immediate successors in A iff $f(a)$ has r immediate successors in B .

8.9 UPPER AND LOWER BOUNDS, SUPREMUM AND INFIMUM

Let B be a subset of a poset $(A, \preceq)$. An element $u \in A$ is called an **upper bound** of B if u succeeds every element of B i.e., $x \preceq u \forall x \in B$ and an element $l \in A$ is called a **lower bound** of B if l precedes every element of B i.e., $l \preceq x \forall x \in B$.

An element $a \in A$ is called a **least upper bound (lub)** of A or **supremum** of A and write $\sup(A)$, if a' is an upper bound of B and $a \preceq a'$ whenever a' is the upper bound of B .

Thus $a = \text{lub}(B)$ if $b \preceq a \forall b \in B$ and if whenever $a' \in A$ is also an upper bound of B , then $a \preceq a'$.

Similarly, an element $a \in A$ is called the **greatest lower bound (glb)** of B or **infimum** of A $\inf(A)$, if a is the lower bound of B and $a' \preceq a$ whenever a' is the lower bound of B . Thus $a = \text{glb}(B)$ if $a \preceq b \forall b \in B$ and if whenever $a' \in A$ is also a lower bound of B then $a' \preceq a$.

NOTE

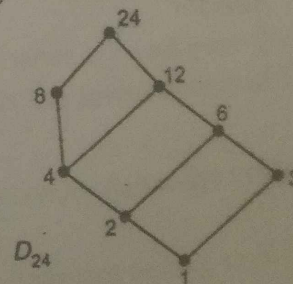
1. A subset B of a poset may or may not have upper or lower bounds.
2. An upper or lower bound may or may not belong to B itself.
3. There can be more than one upper bound and lower bound of a set.
4. The greatest element is always the supremum but the converse is not true.
In fact $a = \sup(B)$ is the greatest element, if and only if $a \in B$.
5. The least element is always the infimum but the converse is not true.
In fact $b = \inf(B)$ is the element iff $b \in B$.

Example 16 Find the greatest lower bound and the least upper bound of the set $\{2, 3, 6\}$ if they exist, in the poset $(D_{24}, \preceq)$.

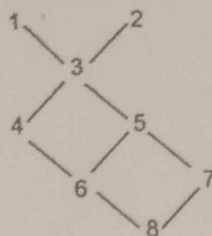
Solution. D_{24} is shown in the figure.

Let $A = \{2, 3, 6\}$ then upper bounds of A are 6, 12, 24 and only lower bound is 1.

Since 1 is the only lower bound, therefore $\inf A = \{1\}$ and $\sup(A) = 6$.



Example 17 Let $S = \{1, 2, \dots, 8\}$ be ordered as



and let $A = \{4, 5, 7\}$. Find, $\inf(A)$ and $\sup(A)$

Solution. The upper bounds of A are 1, 2, 3 and the only lower bound is 8.

Here $\sup(A) = 3$ since 3 precedes the other upper bounds and $\inf(A) = 8$ since 8 is the only lower bound.

Example 18 Let $S = \{1, 2, 3, \dots, 8\}$ be ordered as in previous example and let $A = \{2, 3, 6\}$.

(a) Find the upper and lower bounds of A .

(b) Identify $\sup(A)$ and $\inf(A)$ if either exists.

(c) Repeat (a) and (b) for the subset $B = \{1, 2, 5\}$.

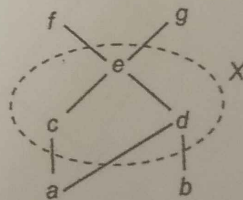
Solution. (a) Upper bounds is 2 and lower bounds are 6 and 8.

(b) Here $\sup(A) = 2$ and $\inf(A) = 16$.

(c) No upper bound for B since no element succeeds both 1 and 2 and lower bounds are 6, 7, 8.

No upper bound so $\sup(B)$ does not exist and no lower bound since no lower bound succeeds both 6 and 7.

Example 19 Let $S = \{a, b, c, d, e, f, g\}$ be ordered as



and let $X = \{c, d, e\}$

(a) Find the upper and lower bounds of X .

(b) Identify $\sup(X)$ and $\inf(X)$ if either exists.

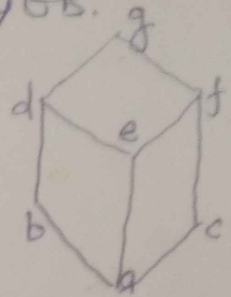
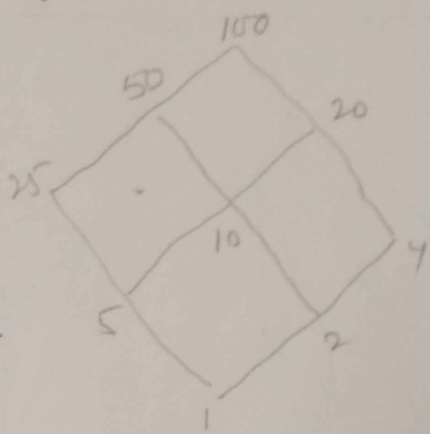
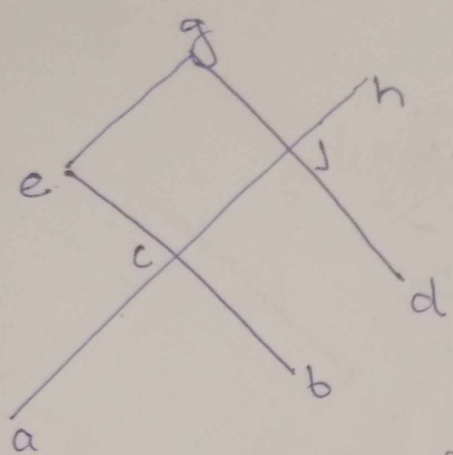
Solution. The elements e, f and g succeed every element of X hence e, f and g are the upper bounds of X . The elements a precedes every element of X , hence it is the lower bound of X . Note that b is not a lower bound since b does not precede c .

(b) Since e precedes both f and g we have $e = \sup(X)$ and a precedes (trivially) every lower bound of X .

We have $a = \inf(X)$. Note that $\sup(X)$ belongs to X but $\inf(X)$ does not belong to X .

upper bound: let B be a subset of a set A . An element $x \in A$ is an upper bound of B if $(x, y) \in \text{Poset } \forall y \in B$

lower bound: let B be a subset of a set A . An element $x \in A$ is a lower bound of B if $(x, y) \in \text{Poset } \forall y \in B$.



$B = \{e, c\}$ $B = \{c, f, d\}$

$B = \{5, 10\}$ $B = \{5, 10, 2, 4\}$

$LB(B) = \{a, b, c\}$ $LB(B) = \emptyset$

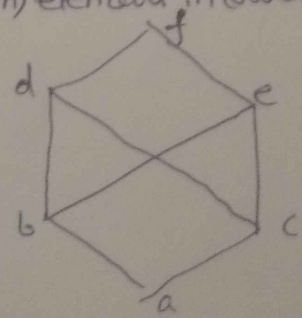
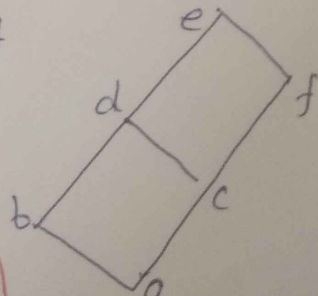
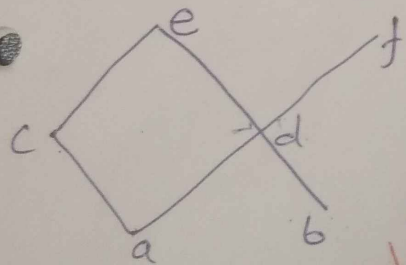
$LB(B) = \{1, 2, 5, 6\}$ $LB(B) = \{1\}$

$UB(B) = \{g, e\}$ $UB(B) = \{g, h, f\}$

$UB(B) = \{100, 50\}$ $UB(B) = \{100, 20\}$

$B = \{d, g\}$ $B = \{e, f\}$
 $LB(B) = \{a, b, c, d\}$
 $UB(B) = \{g\}$
 $LB(B) = \{a, e\}$
 $UB(B) = \{g, f\}$

Least Upper Bound (LUB) (Supremum) (Join) ($\vee$) — Least (minimum) element in Upper bound
Greatest Lower Bound (GLB) (Infimum) (Meet) ($\wedge$) — Greatest (maximum) element in lower bound.



$B = \{c, d\}$, $B = \{a, b\}$, $B = \{e, f\}$
 $UB(B) = e$ $\vdash efd$ $\vdash \emptyset$
 $LB(B) = e$ d $\emptyset$
 $LB(B) = a$ $\emptyset$ adb
 $GLB(B) = a$ $\emptyset$ d

$B = \{a, c, f\}$
 $UB(B) = ef$
 $LB(B) = f$
 $LB(B) = a$
 $GLB(B) = a$

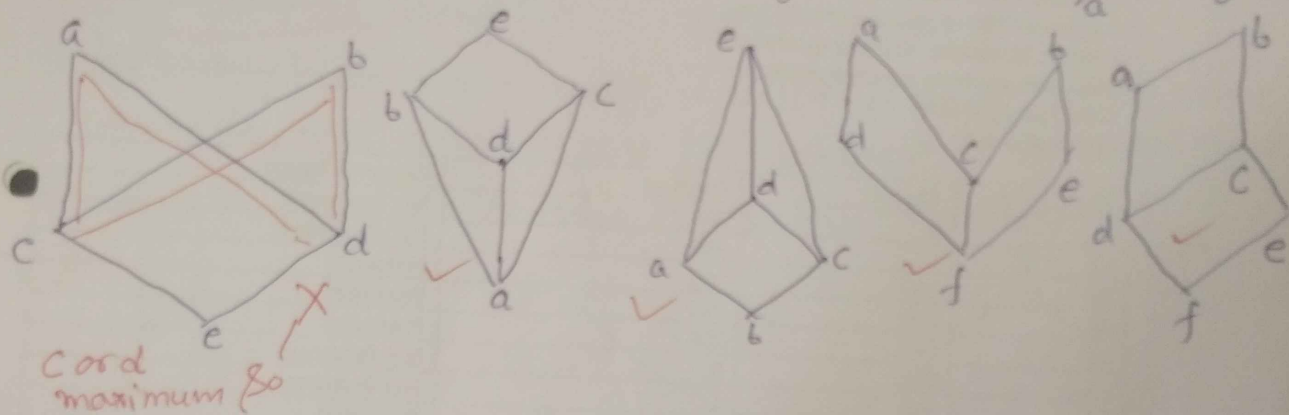
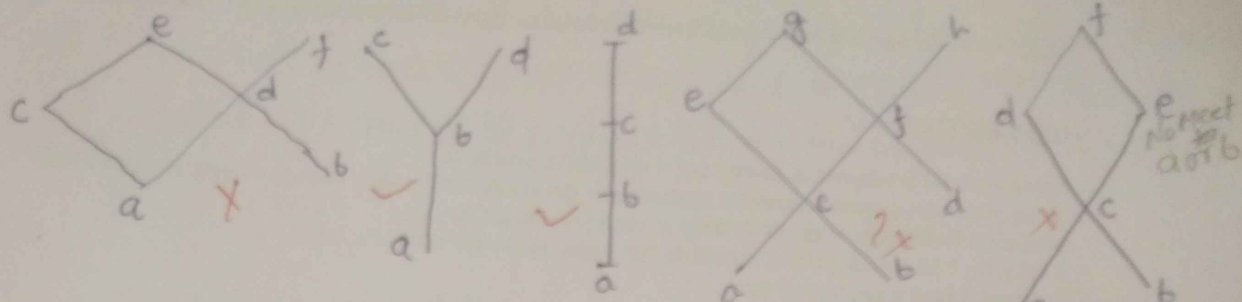
$B = \{d, c\}$
 $UB(B) = ed$
 d
 a, c
 c

$B = \{d, e\}$
 $UB(B) = f$
 $UB(B) = f$
 $LB(B) = b, a, c$
 $GLB(B) = \emptyset$

$B = \{b, c\}$
 dfe
 $\emptyset$ (d, e same level)
 a
 a

as b & c have same level.

Meet-Semi-lattice: - in a poset if GUB/MEET/infimum/ $\wedge$ exist for every pair element then poset is called Meet-Semi-lattice.



LATTICE: A Poset is called lattice if it is both MSL & SSL

→ A Poset $(P, \leq)$ is called a lattice if every two element subset of P has both a least upper bound and a greatest lower bound. i.e if $\text{lub}(x, y)$ and $\text{glb}(x, y)$ exist for every pair of x, y in P . that is

$$x \vee y = \text{lub}(x, y)$$

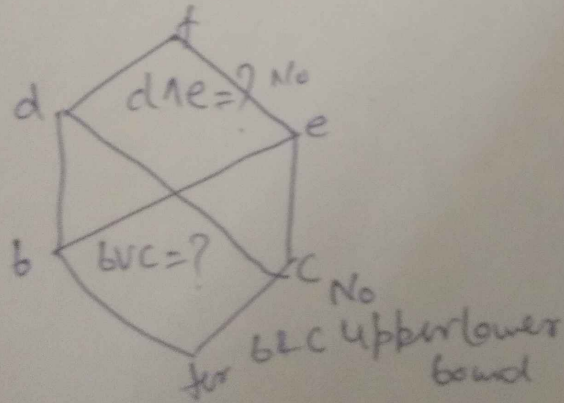
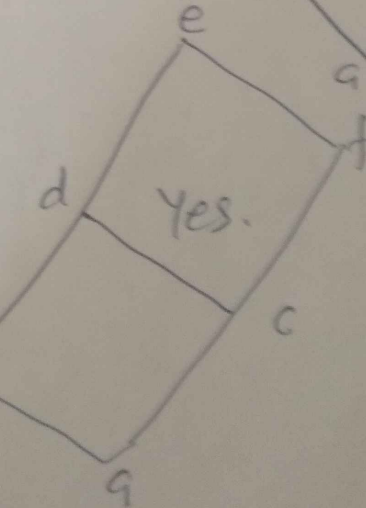
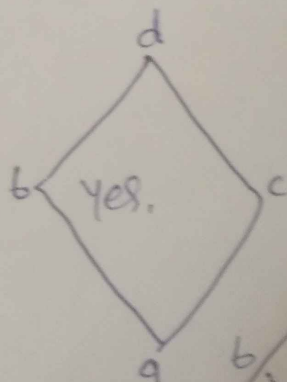
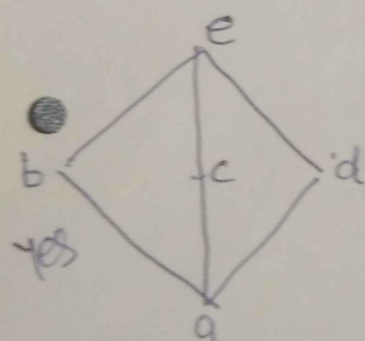
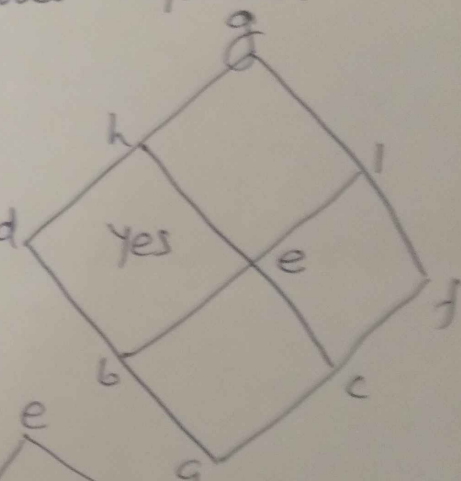
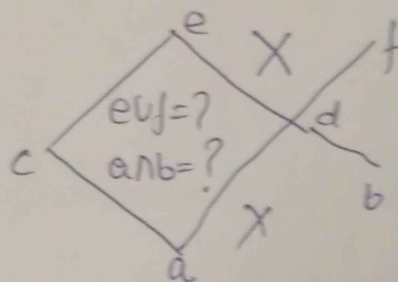
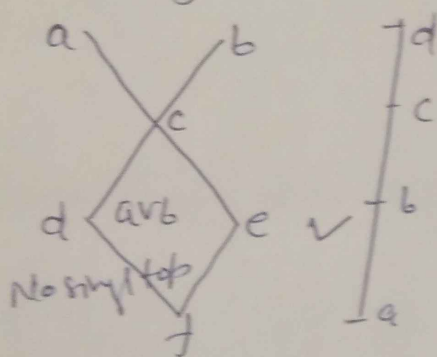
$$(x \wedge y) = \text{glb}(x, y)$$

→ Join means above ↑

Meet means ↓

→ for every pair Join & meet, it will take time, so its time saving to find atleast one pair where it is not Join & meet.

● Single top, single Bottom is essential. / must.



No relation b & c.